Volume Breakout V3E

with Trend-Validator Optimization (score 2-of-3)

BTC / ETH / SOL @ 4h — 2018-01-01 -> 2026-04-01 — $10,000 demo

Metric	V3E Strategy	Buy & Hold
Final equity	**$226,116**	$80,423
Total return	**+2161.2%**	+704.2%
CAGR	**+46.0%**	+28.8%
Sharpe	**1.23**	0.71
Sortino	**1.08**	—
Max drawdown	**-28.7%**	-90.9%
Calmar	**1.60**	0.32

Equity vs BH: 2.81x | DD ratio: 0.32x | Calmar ratio: 5.06x

V2B baseline -> V3E optimized

Metric	V2B baseline	V3E	Delta
CAGR	+42.43%	**+45.96%**	**+3.53**
Sharpe	1.21	**1.23**	**+0.02**
Max drawdown	-31.02%	**-28.69%**	**+2.33**
Calmar	1.37	**1.60**	**+0.23**
Final equity	184,787	**$226,116**	**+$41,329**

Trend validators (entry requires score >= 2 of 3)

```
G1: higher-timeframe 1-Day 200-EMA is rising (macro trend)
G2: ADX(14) > 20 (trend strength, avoid chop)
G3: short-term SMA(50) rising (momentum alignment)
```

Honest walk-forward verification

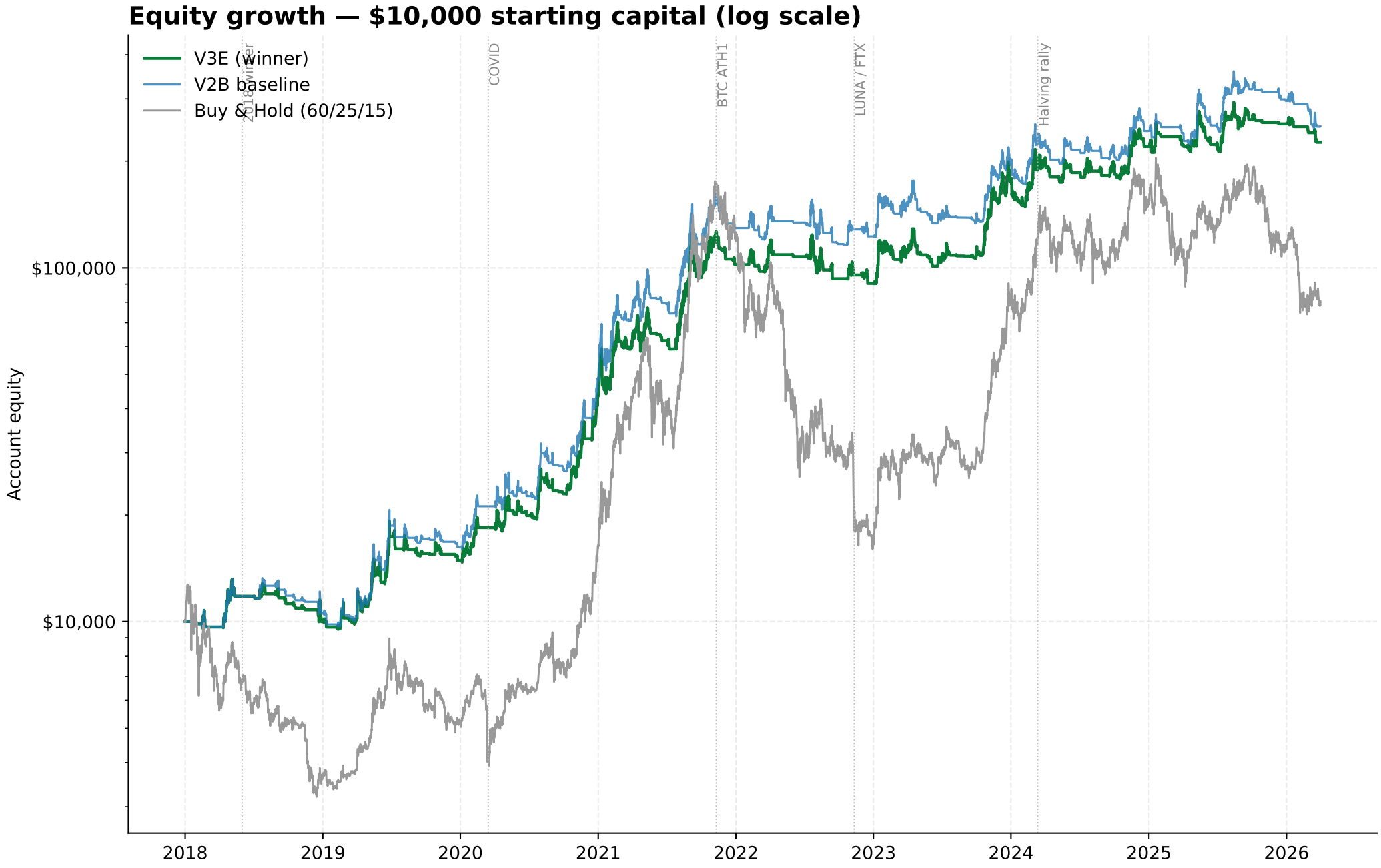
IS 2018-2022: CAGR 55.3% Sharpe 1.36 DD -28.5% Calmar 1.94

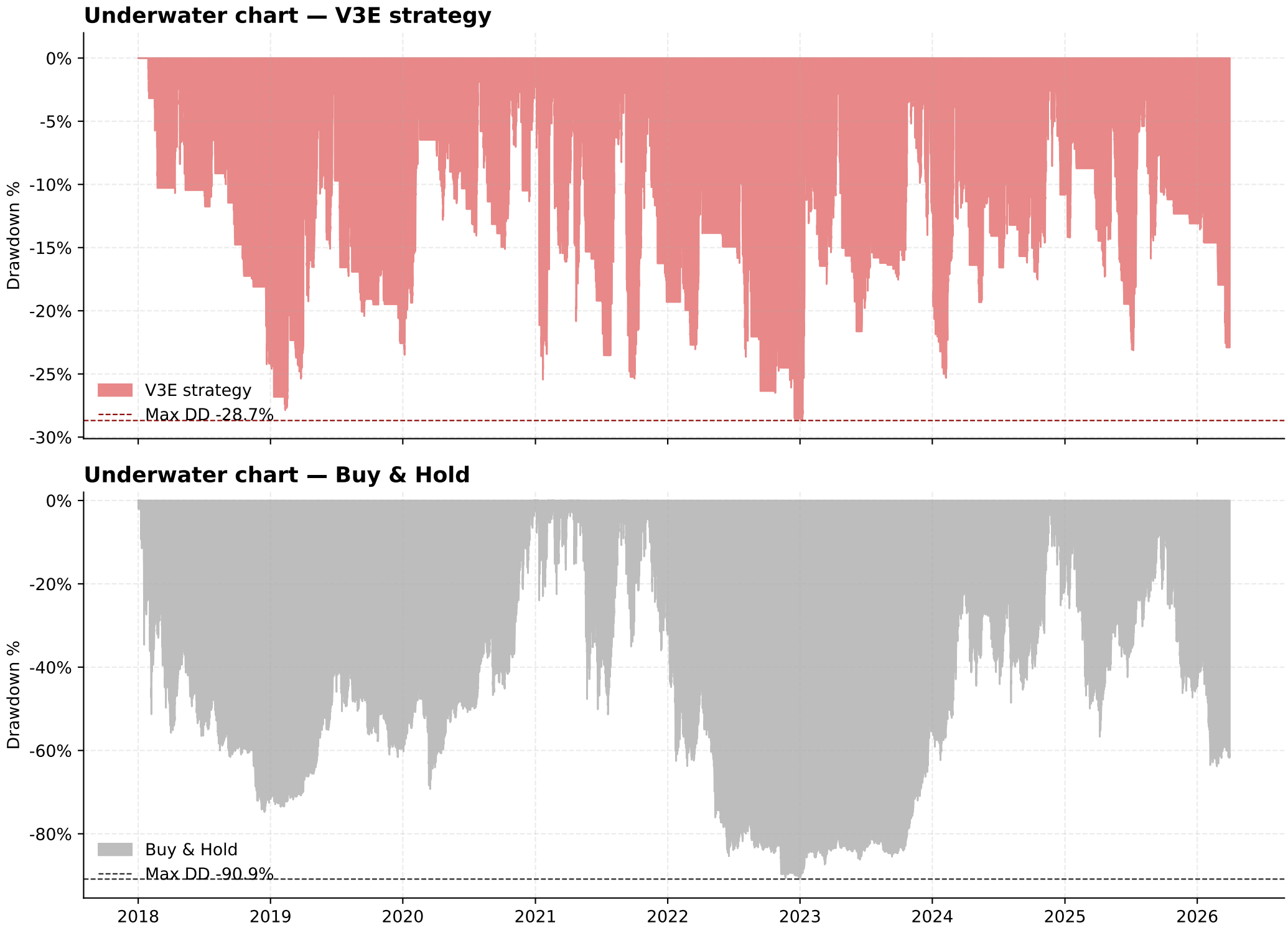
OOS 2023-2026 (unseen): CAGR 23.6% Sharpe 0.91 DD -19.9% Calmar 1.19

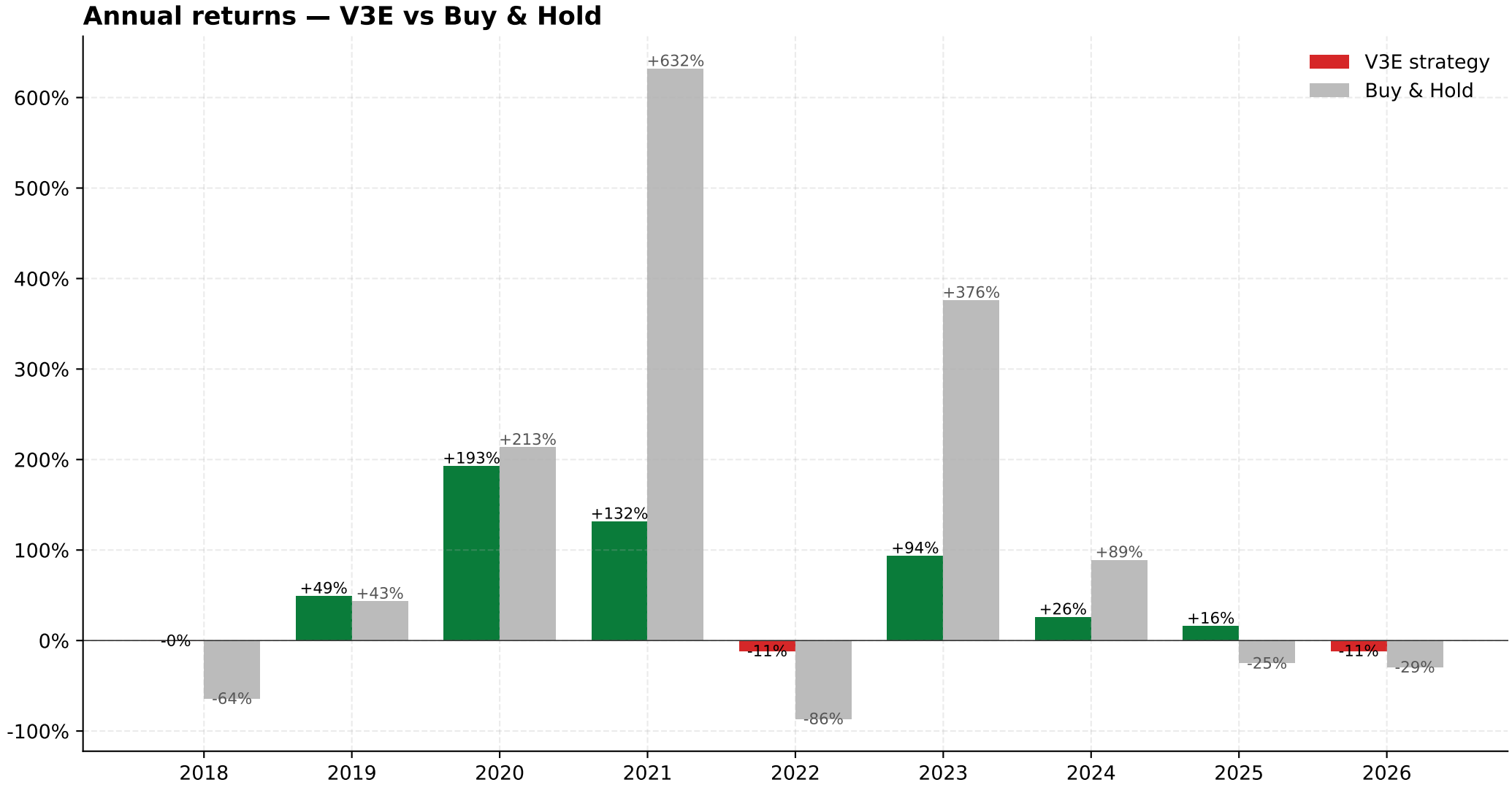
Per-asset contribution

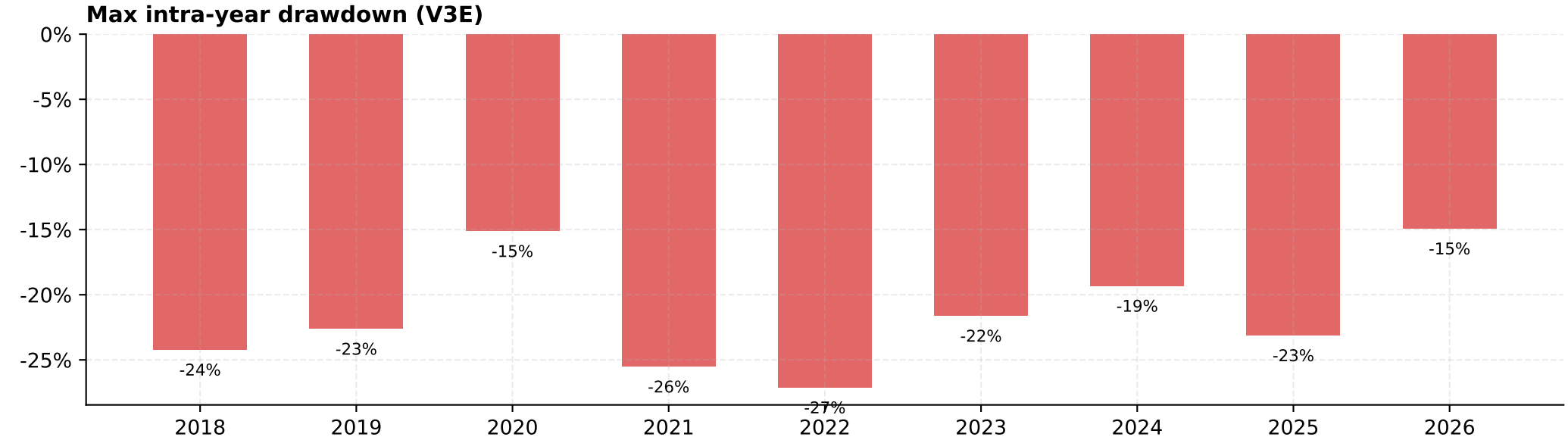
Asset	Init	CAGR	Sharpe	MaxDD	Trades	Win %	BH ret
BTCUSDT	$6,000	+28.2%	0.90	-46.6%	117	32.5%	+409%
ETHUSDT	$2,500	+51.9%	1.16	-39.1%	99	37.4%	+187%
SOLUSDT	$1,500	+111.0%	1.38	-68.8%	79	38.0%	+2747%

Fees: 0.1 % per side + 5 bps slippage. Execution: next bar OPEN after signal close


Equity growth — $10,000 starting capital (log scale)
V3E (winner)
V2B baseline
Buy & Hold (60/25/15)
2018 winter
COVID
BTC ATH1
LUNA / FTX
Halving rally
$100,000
$10,000
Account equity
2018
2019
2020
2021
2022
2023
2024
2025
2026


Underwater chart — V3E strategy
Drawdown %
0%
-5%
-10%
-15%
-20%
-25%
-30%
V3E strategy
Max DD -28.7%
Underwater chart — Buy & Hold
Drawdown %
0%
-20%
-40%
-60%
-80%
Buy & Hold
Max DD -90.9%
2018
2019
2020
2021
2022
2023
2024
2025
2026


Annual returns — V3E vs Buy & Hold
V3E strategy
Buy & Hold
600%
500%
400%
300%
200%
100%
0%
-100%
-0%
-64%
+49%
+43%
+193%
+213%
+132%
+632%
-11%
-86%
+94%
+376%
+26%
+89%
+16%
-25%
-11%
-29%
2018
2019
2020
2021
2022
2023
2024
2025
2026


Max intra-year drawdown (V3E)
0%
-5%
-10%
-15%
-20%
-25%
-24%
-23%
-15%
-26%
-27%
-22%
-19%
-23%
-15%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Trend-validator variants — performance comparison

Variant	Full CAGR	Full Sharpe	Full DD	Full Calmar	OOS CAGR	OOS Sharpe	OOS DD	OOS Calmar	Trades
V2B_base	+42.4%	1.21	-31.0%	1.37	+21.6%	0.89	-27.4%	0.79	352
V3A_htf1d_	+38.2%	1.22	-35.9%	1.06	+24.3%	0.96	-20.6%	1.18	198
V3B_adx_e	+40.5%	1.18	-29.7%	1.36	+19.6%	0.81	-26.7%	0.74	270
V3C_sma_s	+46.2%	1.23	-34.5%	1.34	+19.1%	0.77	-26.5%	0.72	307
V3D_compos	+34.6%	1.24	-34.7%	1.00	+22.8%	0.95	-18.7%	1.22	153
V3E_score	+46.0%	1.23	-28.7%	1.60	+23.6%	0.91	-19.9%	1.19	295

V3E wins on full-period Calmar, full-period MaxDD, OOS Sharpe, and OOS Calmar vs the V2B baseline. V3D (all-3 required) has slightly better OOS Calmar but 25 % fewer trades and lower CAGR — V3E gives a better return-for-risk trade-off.

Rolling 1-year Sharpe ratio — V3E strategy
94.4 % of rolling windows positive | 25th=0.61 75th=1.70
Rolling Sharpe
3
2
1
0
Median 1.25
2019
2020
2021
2022
2023
2024
2025
2026

Strategy Logic — V3E

Breakout core (V2B carried forward):

```
close > highest(high, 30)[1]
AND volume > sma(volume, 20) x 1.3
AND close > sma(close, 150)                  (mid-term regime)
```

Trend validators (ADDED in V3E — score >= 2 of 3):

```
G1: HTF 1-day 200-EMA rising                 (macro regime)
G2: ADX(14) > 20                             (trend strength)
G3: sma(close, 50) rising                    (short-term trend)
```

Exits (whichever triggers first):

```
a) ATR(14) x 4.5 trailing stop               (ratchets up)
b) close < sma(close, 150)                   (regime failsafe)
```

Execution:

```
- Signal on closed bar i -> fill at OPEN of bar i+1 (no look-ahead)
- Fees     : 0.1 % per side (Binance spot)
- Slippage : 5 ticks
- Position : 100 % of sub-account equity, no pyramiding
```

Portfolio allocation (risk-adjusted):

```
BTC 60 %   ETH 25 %   SOL 15 %   of $10,000 demo
```

Caveats:

```
- Long-only. Shorts consistently hurt in the 2018-2026 crypto cycle.
- SOL asset-level DD -68.8 %. Portfolio DD smooths to -28.7 %.
- OOS Sharpe degraded ~31 % vs IS. Expect live 0.8-1.0.
- Single TF (4h). Next direction: multi-TF confirmation + session filter.
```